

TrES-3b

R-2653

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_160607 · created 2026-08-29T23:37:13+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2457546.7306 +/- 0.0012 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.133 +/- 0.03
Transit depth (Rp/Rs)^2	1.77 +/- 0.79 [%]
Orbital Inclination (inc)	82.23 +/- 1.53
Airmass coefficient 1 (a1)	1.206 +/- 0.014
Airmass coefficient 2 (a2)	-0.1717 +/- 0.0057
Scatter in the residuals of the lightcurve fit is	1.13 %
Best Comparison Star	None
Optimal Aperture	4.07
Optimal Annulus	5.07
Transit Duration (day)	0.05 +/- 0.018

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.133 ± 0.03 vs 0.16309 (deviation 0.99σ)
Duration fit vs published	0.05 d vs 0.0567 d (deviation 0.37σ)
Mid-time O–C	2.4 min
Depth SNR	4.4
Residual scatter	1.13 %

Light curve

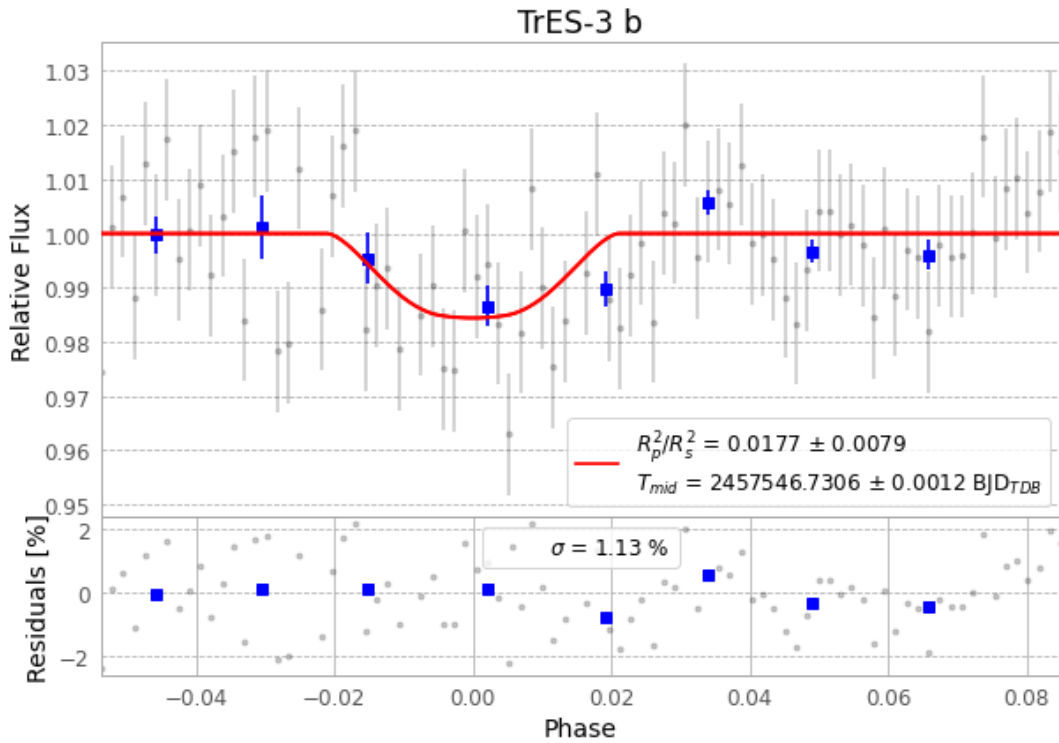


Figure 1 — FinalLightCurve_TrES-3 b_2016-06-06.png (SHA-256 e626b41a94507c14...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [227, 202], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d47769e25fdd6b9c...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

88 FITS frames (58 MB), TRES-3160607034512.FITS → TRES-3160607080612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-160607.log (42ab3a3336cf...), FinalLightCurve_TrES-3 b_2016-06-06.png (e626b41a9450...), FinalLightCurve_TrES-3 b_2016-06-06.csv (e61ae988579f...)

Compute resources

Wall time 118.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 23:37Z.